Binary relations

Outline

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Binary relations

Introduction: Binary relations

- In this section we define **binary relations** as subsets of a Cartesian product
- We will see fundamental examples: "less than", "less than or equal to", "equal to", "is a subset of", congruence modulo n, multiples and divisors
- We will learn to visualize a relation with a **table**
- Relations allow us to formalize the concepts of equivalence, of order and of function

Binary relation

Definition 1: of binary relation

*Given two sets A and B, a **binary relation** R is a subset of the Cartesian product $A \times B$*

- When we say that R is a binary relation on a single set A, we mean that R is a subset of $A \times A$.
- A binary relation will simply be called a relation (however, non-binary relations also exist).
- If $(a, b) \in R$ we also write $a\,R\,b$ and we say that a is **related** to b.

Example (1/2)

- The relation "less than" ("$<$") on the natural numbers is the set:
$$R_< = \Big\{(a,b) : a,b \in \mathbb{N} \ \text{ and } \ a < b\Big\}.$$
- The relation "less than or equal to" ("$\leq$") on the natural numbers is the set:
$$R_\leq = \Big\{(a,b) : a,b \in \mathbb{N} \ \text{ and } \ a \leq b\Big\}.$$
- The relation "equal to" ("$=$") on the natural numbers is the set:
$$R_= = \Big\{(a,b) : a,b \in \mathbb{N} \ \text{ and } \ a = b\Big\}.$$

Example (2/2)

- The relation "is a subset of" $R_{\subseteq}$ on the power set of the natural numbers (denoted by $2^{\mathbb{N}}$) is the set:

$$R_{\subseteq} = \left\{ (A, B) : A, B \in 2^{\mathbb{N}} \text{ and } A \subseteq B \right\}.$$

Table of a relation

A relation R on a set of natural numbers can be visualized with a **table**:

in row a and column b we write the pair (a, b) if $(a, b) \in R$, and the cell is left empty if $(a, b) \notin R$.

Example: tables of $R_<$ and of $R_=$

The relation $R_<$ (rows and columns from 0 to 4):

	0	1	2	3	4	...
0		$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	
1			$(1,2)$	$(1,3)$	$(1,4)$	
2				$(2,3)$	$(2,4)$	
3					$(3,4)$	
4						
...						

The relation $R_=$:

	0	1	2	3	4	...
0	$(0,0)$					
1		$(1,1)$				
2			$(2,2)$			
3				$(3,3)$		
4					$(4,4)$	
...						

Example: table of $R_{\leq}$

The relation $R_{\leq}$ contains the pairs of $R_{<}$ and those of $R_{=}$:

	0	1	2	3	4	...
0	$(0,0)$	$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	
1		$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	
2			$(2,2)$	$(2,3)$	$(2,4)$	
3				$(3,3)$	$(3,4)$	
4					$(4,4)$	
...						

Congruence modulo n

Definition 2: of congruence modulo n

*Given a natural number $n \geq 1$, two natural numbers a and b are **congruent modulo** n, written $a \equiv b \pmod{n}$, if their difference is an integer multiple of n:*

$$a \equiv b \pmod{n} \iff \exists q \in \mathbb{Z} \text{ such that } b - a = q\,n.$$

Multiples and divisors

Definition 3: of multiple and divisor

*Given two natural numbers a and b, we say that a is a **multiple** of b if there exists $k \in \mathbb{N}$ such that $a = k\,b$. We say that a is a **divisor** of b if there exists $k \in \mathbb{N}$ such that $b = k\,a$.*

- Hence a is a divisor of b if and only if b is a multiple of a.
- 0 is a multiple of every natural number ($0 = 0 \cdot b$) and 0 is a divisor only of itself (if $b = k \cdot 0$ then $b = 0$).

Example: congruence modulo 3

The relation "congruent modulo n" on the natural numbers is the set:

$$R_{\equiv_n} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } a \equiv b \pmod{n} \right\}.$$

With $n = 3$ the table is:

	0	1	2	3	4	5	6	...
0	$(0,0)$			$(0,3)$			$(0,6)$	
1		$(1,1)$			$(1,4)$			
2			$(2,2)$			$(2,5)$		
3	$(3,0)$			$(3,3)$			$(3,6)$	
4		$(4,1)$			$(4,4)$			
5			$(5,2)$			$(5,5)$		
6	$(6,0)$			$(6,3)$			$(6,6)$	
...								

Example: "is a multiple of"

$$R_{\text{mul}} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } \exists k \in \mathbb{N} \text{ such that } a = k\, b \right\}$$

	0	1	2	3	4	5	6	...
0	$(0,0)$	$(0,1)$	$(0,2)$	$(0,3)$	$(0,4)$	$(0,5)$	$(0,6)$	
1		$(1,1)$						
2		$(2,1)$	$(2,2)$					
3		$(3,1)$		$(3,3)$				
4		$(4,1)$	$(4,2)$		$(4,4)$			
5		$(5,1)$				$(5,5)$		
6		$(6,1)$	$(6,2)$	$(6,3)$			$(6,6)$	
...								

Example: "is a divisor of"

$$R_{\text{div}} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } \exists k \in \mathbb{N} \text{ such that } b = k\, a \right\}$$

Since $(a, b) \in R_{\text{div}}$ if and only if $(b, a) \in R_{\text{mul}}$, the table is obtained from that of R_{mul} by swapping the elements of each pair:

	0	1	2	3	4	5	6	...
0	(0, 0)							
1	(1, 0)	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)	
2	(2, 0)		(2, 2)		(2, 4)		(2, 6)	
3	(3, 0)			(3, 3)			(3, 6)	
4	(4, 0)				(4, 4)			
5	(5, 0)					(5, 5)		
6	(6, 0)						(6, 6)	
...								

Example: "difference of one"

$$R^1_{ab} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } a = b - 1 \right\}$$

	0	1	2	3	4	5	6	...
0		$(0,1)$						
1			$(1,2)$					
2				$(2,3)$				
3					$(3,4)$			
4						$(4,5)$		
5							$(5,6)$	
6								
...								

Example: "the sum is even"

$$R_{\text{even}} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } \exists m \in \mathbb{N} \text{ such that } a + b = 2\,m \right\}$$

	0	1	2	3	4	5	6	...
0	(0, 0)		(0, 2)		(0, 4)		(0, 6)	
1		(1, 1)		(1, 3)		(1, 5)		
2	(2, 0)		(2, 2)		(2, 4)		(2, 6)	
3		(3, 1)		(3, 3)		(3, 5)		
4	(4, 0)		(4, 2)		(4, 4)		(4, 6)	
5		(5, 1)		(5, 3)		(5, 5)		
6	(6, 0)		(6, 2)		(6, 4)		(6, 6)	
...								

Example: "is a subset of"

The table of $R_{\subseteq}$ restricted to the subsets of $\{0,1,2\}$:

	$\emptyset$	$\{0\}$	$\{1\}$	$\{2\}$	$\{0,1\}$	$\{0,2\}$	$\{1,2\}$	$\{0,1,2\}$
$\emptyset$	$(\emptyset,\emptyset)$	$(\emptyset,\{0\})$	$(\emptyset,\{1\})$	$(\emptyset,\{2\})$	$(\emptyset,\{0,1\})$	$(\emptyset,\{0,2\})$	$(\emptyset,\{1,2\})$	$(\emptyset,\{0,1,2\})$
$\{0\}$		$(\{0\},\{0\})$			$(\{0\},\{0,1\})$	$(\{0\},\{0,2\})$		$(\{0\},\{0,1,2\})$
$\{1\}$			$(\{1\},\{1\})$		$(\{1\},\{0,1\})$		$(\{1\},\{1,2\})$	$(\{1\},\{0,1,2\})$
$\{2\}$				$(\{2\},\{2\})$		$(\{2\},\{0,2\})$	$(\{2\},\{1,2\})$	$(\{2\},\{0,1,2\})$
$\{0,1\}$					$(\{0,1\},\{0,1\})$			$(\{0,1\},\{0,1,2\})$
$\{0,2\}$						$(\{0,2\},\{0,2\})$		$(\{0,2\},\{0,1,2\})$
$\{1,2\}$							$(\{1,2\},\{1,2\})$	$(\{1,2\},\{0,1,2\})$
$\{0,1,2\}$								$(\{0,1,2\},\{0,1,2\})$

n-ary relation

Definition 4: of n-ary relation

*Given n sets $A_1, A_2, \ldots, A_n$, an n-**ary relation** R is a subset of the Cartesian product $A_1 \times A_2 \times \cdots \times A_n$.*

- Binary relations are the n-ary relations with $n = 2$.

Summary

- A **binary relation** between A and B is a subset of $A \times B$; an n-ary relation is a subset of $A_1 \times A_2 \times \cdots \times A_n$
- Examples: $R_<$, $R_\leq$, $R_=$ on the naturals, $R_\subseteq$ on $2^{\mathbb{N}}$
- Congruence modulo n: $b - a = q\,n$ with $q \in \mathbb{Z}$
- Multiple: $a = k\,b$; divisor: $b = k\,a$, with $k \in \mathbb{N}$
- A relation on the naturals can be visualized with a **table**

Properties of relations

Introduction: Properties of relations

- In this section we introduce four properties: **reflexive**, **symmetric**, **transitive** and **antisymmetric**
- For each one we will see how to prove that a relation has it and how to show, with a counterexample, that it does not
- Combining these properties we will define equivalence relations and order relations

Reflexive, symmetric and transitive properties

- A relation $R \subseteq A \times A$ is **reflexive** if:

$$\forall a \in A, \qquad (a, a) \in R$$

- A relation $R \subseteq A \times A$ is **symmetric** if:

$$\forall a, b \in A, \qquad (a, b) \in R \quad \Rightarrow \quad (b, a) \in R$$

- A relation $R \subseteq A \times A$ is **transitive** if:

$$\forall a, b, c \in A, \qquad (a, b) \in R \quad \text{and} \quad (b, c) \in R \quad \Rightarrow \quad (a, c) \in R$$

Antisymmetric property

- A relation $R \subseteq A \times A$ is **antisymmetric** if:

$$\forall a, b \in A, \quad (a, b) \in R \quad \text{and} \quad (b, a) \in R \quad \Rightarrow \quad a = b$$

Observation 1: equivalent form of antisymmetry

A relation $R \subseteq A \times A$ is antisymmetric if and only if:

$$\forall a, b \in A, \quad (a, b) \in R \quad \text{and} \quad a \neq b \quad \Rightarrow \quad (b, a) \notin R$$

Proof

Fix $a, b \in A$. An implication is false only when the antecedent is true and the consequent is false.

- "$(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$" is false only when

$$(a, b) \in R, \quad (b, a) \in R, \quad a \neq b.$$

- "$(a, b) \in R$ and $a \neq b \Rightarrow (b, a) \notin R$" is false only when

$$(a, b) \in R, \quad a \neq b, \quad (b, a) \in R.$$

The two implications are false in exactly the same cases, hence one holds for all $a, b \in A$ if and only if the other does. □

Example

- The relation $R_{\leq}$ is reflexive, but $R_{<}$ is not.
- The relations $R_{<}$ and $R_{\leq}$ are not symmetric.
- The relations $R_{<}$ and $R_{\leq}$ are transitive, but for example the relation:

$$R^1_{ab} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } a = b - 1 \right\}$$

is not, e.g., $(3, 4) \in R^1_{ab}$ and $(4, 5) \in R^1_{ab}$ but $(3, 5) \notin R^1_{ab}$.

Let us now see in detail, for each property, how to prove that a relation has it and how to show, with a counterexample, that it does not.

Example: reflexive property

- $R_=$ is reflexive, since

$$a = a, \qquad \forall a \in \mathbb{N}.$$

- $R_\leq$ is reflexive, since

$$a \leq a, \qquad \forall a \in \mathbb{N}.$$

- $R_<$ is not reflexive: a counterexample is $a = 1$, since $1 < 1$ is false and hence $(1, 1) \notin R_<$.

Example: symmetric property

- $R_=$ is symmetric, since

$$a = b \quad \Rightarrow \quad b = a, \qquad \forall a, b \in \mathbb{N}.$$

- $R_<$ and $R_\leq$ are not symmetric: a counterexample is the pair $(1, 2)$, since $1 < 2$ and $1 \leq 2$, but $2 < 1$ and $2 \leq 1$ are false.

  Hence $(1, 2) \in R_<$ but $(2, 1) \notin R_<$, and $(1, 2) \in R_\leq$ but $(2, 1) \notin R_\leq$.

Example: transitive property

- $R_=$ is transitive, since

$$a = b \quad \text{and} \quad b = c \quad \Rightarrow \quad a = c, \qquad \forall a, b, c \in \mathbb{N}.$$

  The same holds for $R_<$ and $R_\leq$: if $a < b$ and $b < c$ then $a < c$, and if $a \leq b$ and $b \leq c$ then $a \leq c$.

- R^1_{ab} is not transitive: $(3, 4) \in R^1_{ab}$ and $(4, 5) \in R^1_{ab}$, while $(3, 5) \notin R^1_{ab}$ because $3 \neq 5 - 1$.

Example: antisymmetric property (1/2)

- $R_{\leq}$ is antisymmetric, since $a \leq b$ and $b \leq a$ imply $a = b$.
- $R_{=}$ is antisymmetric, since

$$a = b \quad \text{and} \quad b = a \quad \Rightarrow \quad a = b, \qquad \forall a, b \in \mathbb{N}.$$

- $R_{\equiv_3}$ is not antisymmetric: $(0, 3) \in R_{\equiv_3}$ (because $3 - 0 = 1 \cdot 3$) and $(3, 0) \in R_{\equiv_3}$ (because $0 - 3 = (-1) \cdot 3$), while $0 \neq 3$.

Example: antisymmetric property (2/2)

- $R_<$ is antisymmetric. There are no two natural numbers a and b with $a < b$ and $b < a$ (otherwise we would have $a < a$): the antecedent of the implication is never true, and hence the implication is true for all $a, b \in \mathbb{N}$.
- With the equivalent form of antisymmetry this can be seen directly:

$$a < b \quad \text{and} \quad a \neq b \quad \Rightarrow \quad b \not< a, \qquad \forall a, b \in \mathbb{N}.$$

Summary

- **Reflexive**: $(a, a) \in R$ for every a
- **Symmetric**: $(a, b) \in R \Rightarrow (b, a) \in R$
- **Transitive**: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$
- **Antisymmetric**: $(a, b), (b, a) \in R \Rightarrow a = b$, or equivalently $(a, b) \in R,\ a \neq b \Rightarrow (b, a) \notin R$
- To show that a property does not hold a **counterexample** suffices
- $R_<$ is antisymmetric (vacuously), but not reflexive

Representation with directed graphs

Introduction: Representation with directed graphs

- In this section we represent relations with **directed graphs**: vertices and arrows
- We will draw the graphs of the relations seen so far
- We will learn to read the properties of a relation directly on its graph
- We will see graphs satisfying two properties but not the third

Directed graph of a relation

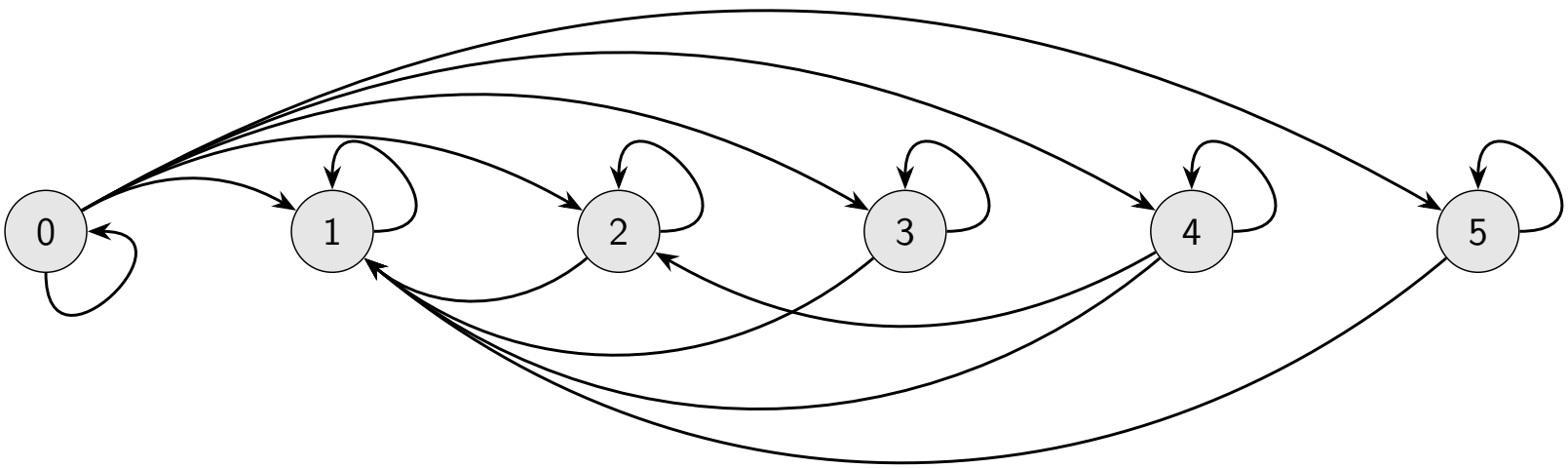
Relations can be represented by a **directed graph**, where the **vertices** are the elements of the set on which the relation R is defined and an **arc** (a, b), drawn as an arrow from a to b, means that $(a, b) \in R$.

An arc (a, a) is called a **loop**.

Example: "is a multiple of"

The directed graph of the relation "is a multiple of" on the set $\{0, 1, \ldots, 5\}$:

$$R = \Big\{ (a, b) : a, b \in \{0, 1, \ldots, 5\} \ \text{ and } \ \exists k \in \mathbb{N} : \ a = k\, b \Big\}$$

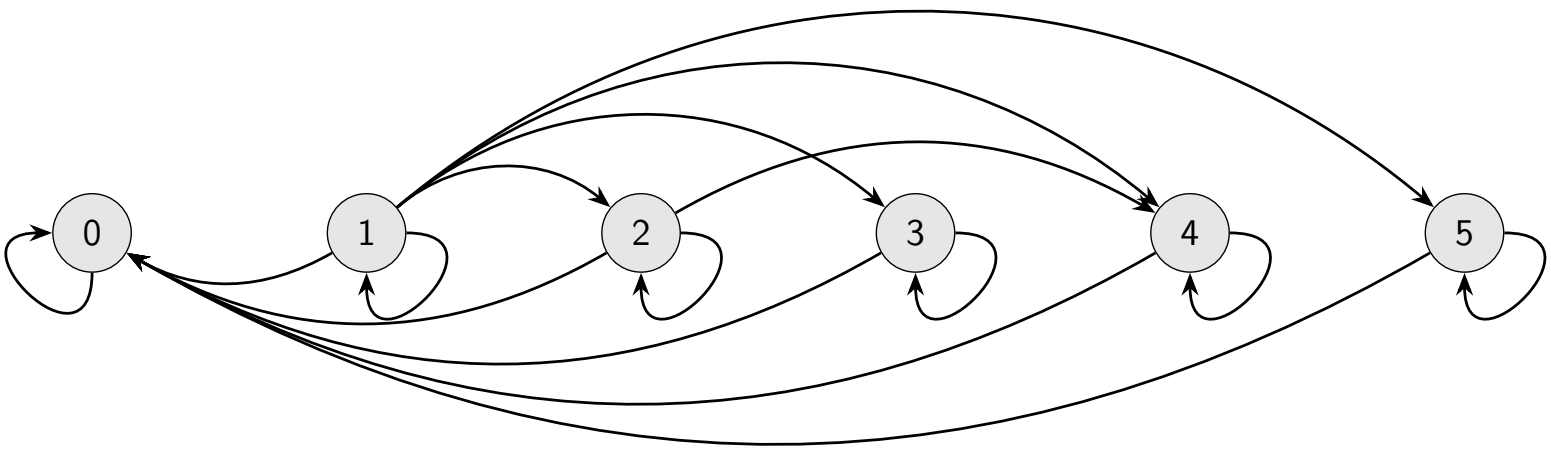


Example: "is a divisor of"

The directed graph of the relation "is a divisor of" on the set $\{0, 1, \ldots, 5\}$:

$$R = \left\{ (a, b) : a, b \in \{0, 1, \ldots, 5\} \text{ and } \exists k \in \mathbb{N} : \ b = k\, a \right\}$$

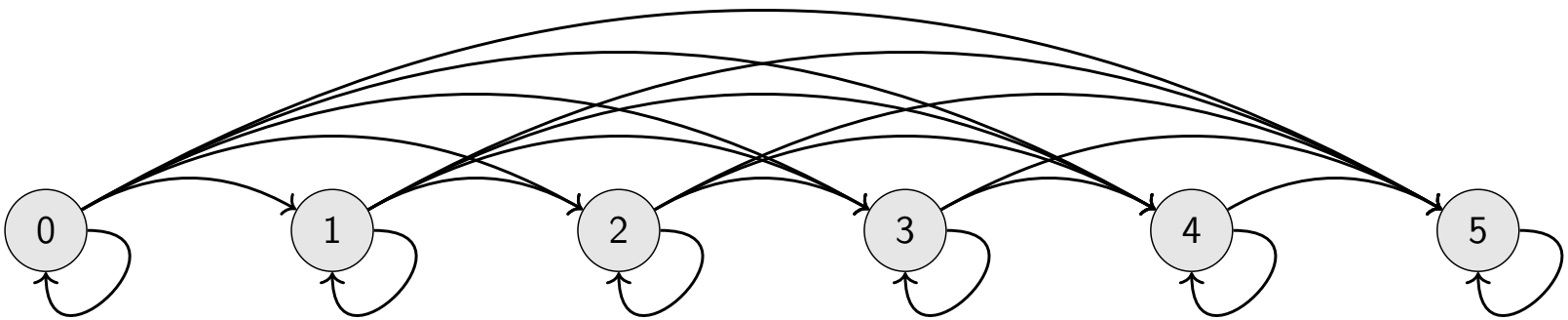
It has the same arcs as the previous graph, traversed in the opposite direction.



Example: "less than or equal to"

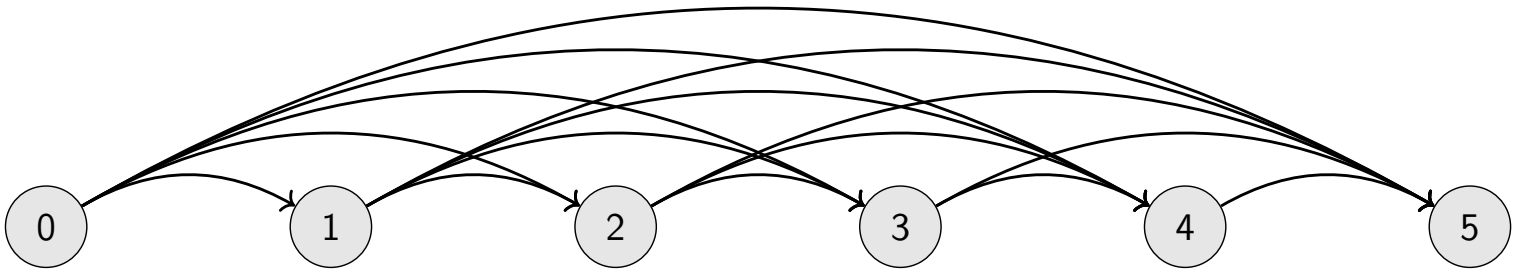
The directed graph of the relation "less than or equal to" on the set $\{0, 1, \ldots, 5\}$:

$$R = \Big\{ (a, b) : a, b \in \{0, 1, \ldots, 5\} \quad \text{and} \quad a \leq b \Big\}$$



Example: "less than" and "equal to"

The graph of "less than" on $\{0, 1, \ldots, 5\}$ is obtained from the previous one by removing the loops:



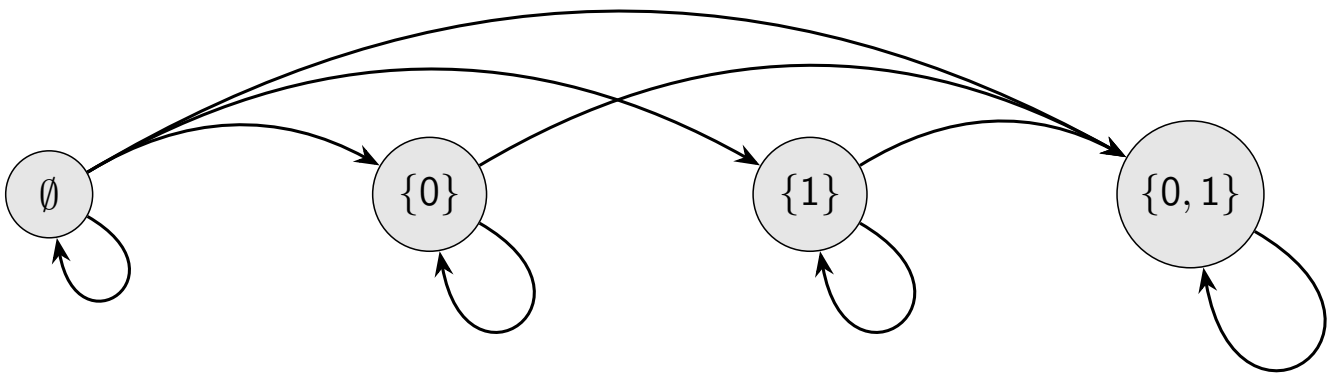
The graph of "equal to" on $\{0, 1, \ldots, 5\}$ has only the loops:



Example: "is a subset of"

The directed graph of the relation "is a subset of" on the subsets of $\{0, 1\}$:

$$R = \left\{ (A, B) : A, B \in 2^{\{0,1\}} \quad \text{and} \quad A \subseteq B \right\}$$



The properties on the graph

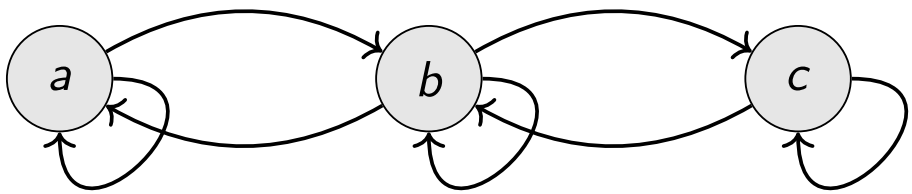
- **Reflexive** if and only if every vertex a has the loop (a, a)
- **Symmetric** if and only if, for every pair of vertices a and b, if there is the arc from a to b then there is also the arc from b to a
- **Transitive** if and only if, for every triple of vertices a, b and c, if there are the arc from a to b and the arc from b to c then there is also the arc from a to c
- **Antisymmetric** if and only if, for every pair of distinct vertices a and b, if there is the arc from a to b then there is no arc from b to a

The properties on the graphs seen

- In the graphs of "is a multiple of", "is a divisor of", "less than or equal to" and "equal to" every vertex has the loop: they are **reflexive** relations
- In the graph of "less than" no vertex has the loop
- In none of these graphs are there two arcs in opposite directions between two distinct vertices: they are **antisymmetric** relations

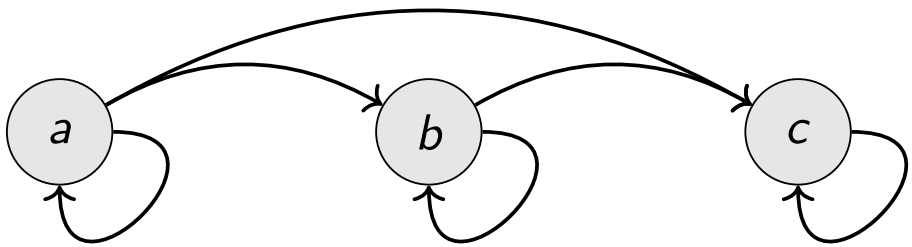
Example: graphs satisfying only two properties (1/3)

The relation $\{(a,a),(b,b),(c,c),(a,b),(b,a),(b,c),(c,b)\}$ on the set $\{a,b,c\}$ is reflexive and symmetric, but not transitive: there are the arcs (a,b) and (b,c), but not the arc (a,c).



Example: graphs satisfying only two properties (2/3)

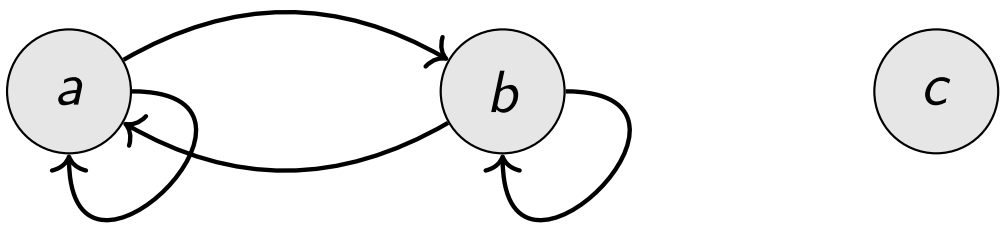
The relation $\{(a,a),(b,b),(c,c),(a,b),(a,c),(b,c)\}$ on the set $\{a,b,c\}$ is reflexive and transitive, but not symmetric: there is the arc (a,b), but not the arc (b,a).



Example: graphs satisfying only two properties (3/3)

The relation $\{(a, a), (a, b), (b, a), (b, b)\}$ on the set $\{a, b, c\}$ is symmetric and transitive, but not reflexive: the loop (c, c) is missing.

For transitivity: from (a, b) and (b, a) follows (a, a), from (b, a) and (a, b) follows (b, b), and the pairs containing a loop give an arc already present.



Symmetric and transitive relations

Observation 2: symmetric and transitive relations

Let $R \subseteq A \times A$ be a symmetric and transitive relation. If an element $a \in A$ appears in at least one pair of R, then $(a, a) \in R$.

- Hence a symmetric and transitive relation fails to be reflexive only if some element of A appears in no pair of the relation, like the element c of the last graph.

Proof

If a appears in a pair of R, there exists $b \in A$ such that

$$(a, b) \in R \quad \text{or} \quad (b, a) \in R.$$

By symmetry, in both cases we have

$$(a, b) \in R \quad \text{and} \quad (b, a) \in R.$$

By transitivity, from $(a, b) \in R$ and $(b, a) \in R$ follows

$$(a, a) \in R.$$

□

Summary

- A relation is represented by a **directed graph**: arc (a, b) if and only if $(a, b) \in R$
- The properties can be read on the graph: loops (reflexive), arcs in both directions (symmetric), "shortcuts" (transitive), never both directions between distinct vertices (antisymmetric)
- The reflexive, symmetric and transitive properties are independent: two of them can hold without the third
- Symmetric and transitive $\Rightarrow (a, a) \in R$ for every a appearing in a pair

Equivalence relations

Introduction: Equivalence relations

- In this section we study **reflexive, symmetric and transitive** relations: the equivalence relations
- They formalize the idea of "having something in common" (the same remainder, the same parity, ...)
- We will see **equivalence classes** and the link with the **partitions** of a set
- Fundamental examples: congruence modulo n, even and odd numbers

Equivalence relation

Definition 5: of equivalence relation

A reflexive, symmetric and transitive relation is an ***equivalence relation****.*

Example

- $R_=$ is an equivalence relation, because it is reflexive, symmetric and transitive.
- $R_<$ is not an equivalence relation, because it is not reflexive (nor symmetric).
- $R_\leq$ is not an equivalence relation, because it is not symmetric.

Example: congruence modulo n (1/2)

Given a natural number $n \geq 1$, the relation

$$R_{\equiv_n} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } a \equiv b \pmod{n} \right\}$$

is an equivalence relation. Recall that $(a, b) \in R_{\equiv_n}$ if and only if there exists $q \in \mathbb{Z}$ such that $b - a = q\, n$.

- Reflexive: for every $a \in \mathbb{N}$ we have $a - a = 0 \cdot n$, with $0 \in \mathbb{Z}$, hence $(a, a) \in R_{\equiv_n}$.

Example: congruence modulo n (2/2)

- Symmetric: if $b - a = q\,n$ with $q \in \mathbb{Z}$, then

$$a - b = (-q)\,n, \qquad \text{with} \quad -q \in \mathbb{Z},$$

  hence $(b, a) \in R_{\equiv_n}$.
- Transitive: if $b - a = q\,n$ and $c - b = p\,n$ with $q, p \in \mathbb{Z}$, then

$$c - a = (c - b) + (b - a) = p\,n + q\,n = (q + p)\,n,$$

  with $q + p \in \mathbb{Z}$, hence $(a, c) \in R_{\equiv_n}$.

Example: the sum is even

The relation

$$R_{\text{even}} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } \exists m \in \mathbb{N} \text{ such that } a + b = 2\,m \right\}$$

is an equivalence relation.

- Reflexive: $a + a = 2\,a$, with $a \in \mathbb{N}$.
- Symmetric: if $a + b = 2\,m$, then also $b + a = a + b = 2\,m$.
- Transitive: if $a + b = 2\,p$ and $b + c = 2\,q$ with $p, q \in \mathbb{N}$, then

$$(a + b) + (b + c) = 2\,p + 2\,q \quad \Rightarrow \quad a + c = 2\,(p + q - b).$$

$p + q - b$ is a nonnegative integer, because
$2\,(p + q - b) = a + c \geq 0$.

Equivalence class

Definition 6: of equivalence class

Given an equivalence relation R on a set A, the ***equivalence class*** *of an element $a \in A$ is the set*

$$[a] = \left\{ b \in A : (a, b) \in R \right\}.$$

Lemma 1: related elements have the same class

Let R be an equivalence relation on a set A. For all $a, b \in A$:

$$(a, b) \in R \quad \Rightarrow \quad [a] = [b].$$

Proof

Let $(a, b) \in R$. We show the two inclusions.

- $[b] \subseteq [a]$: if $c \in [b]$, then $(b, c) \in R$. From $(a, b) \in R$ and $(b, c) \in R$, by transitivity, follows $(a, c) \in R$, that is $c \in [a]$.
- $[a] \subseteq [b]$: by symmetry $(b, a) \in R$. If $c \in [a]$, then $(a, c) \in R$, and from $(b, a) \in R$ and $(a, c) \in R$, by transitivity, follows $(b, c) \in R$, that is $c \in [b]$.

Hence $[a] = [b]$. □

Elements of the same class

- Any two elements of the same class are related to each other: if $b, c \in [a]$, then

$$(a, b) \in R \quad \text{and} \quad (a, c) \in R;$$

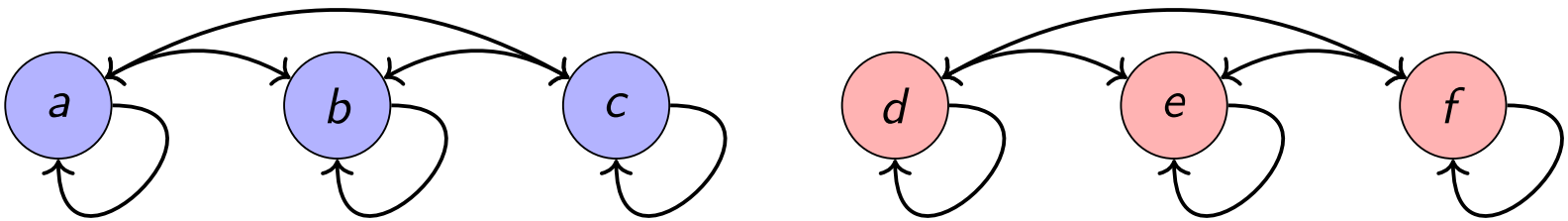
by symmetry $(b, a) \in R$ and by transitivity $(b, c) \in R$.

- An equivalence class therefore gathers elements that are all related to each other.

Example: two equivalence classes

The directed graph of an equivalence relation on the set $\{a, b, c, d, e, f\}$ with two equivalence classes, $\{a, b, c\}$ (in blue) and $\{d, e, f\}$ (in red).

A double-headed arrow between x and y represents the two arcs (x, y) and (y, x).



Partition

Definition 7: of partition

*A **partition** of a set A is a family $\mathcal{S}$ of subsets of A such that:*

- $S \neq \emptyset$ *for every* $S \in \mathcal{S}$;
- $S \cap T = \emptyset$ *for all* $S, T \in \mathcal{S}$ *with* $S \neq T$;
- $\bigcup_{S \in \mathcal{S}} S = A$, *that is, every* $a \in A$ *belongs to at least one* $S \in \mathcal{S}$.

- The family can be finite, $\mathcal{S} = \{S_1, S_2, \ldots, S_n\}$, or infinite.
- Every element of A belongs to **one and only one** set of the partition.

From classes to partition

Proposition 1: equivalence classes form a partition

Let R be an equivalence relation on a set A. Then the family of the equivalence classes

$$\mathcal{S} = \left\{ [a] : a \in A \right\}$$

is a partition of A.

- The proof checks the three conditions of the definition of partition; the second one follows from the fact that two classes are either equal or disjoint.

Proof

Every class $[a]$ is a subset of A by definition.

- Nonempty classes: by reflexivity $(a, a) \in R$, hence $a \in [a]$.
- Union equal to A: every $a \in A$ belongs to the class $[a]$, and every class is contained in A.
- Distinct classes are disjoint: let $c \in [a] \cap [b]$. Then

$$(a, c) \in R \quad \text{and} \quad (b, c) \in R.$$

By symmetry $(c, b) \in R$; by transitivity, from (a, c) and (c, b) follows $(a, b) \in R$, and by the Lemma $[a] = [b]$.

Hence two distinct classes have no element in common. □

From partition to classes

Proposition 2: every partition defines an equivalence relation

Let $\mathcal{S}$ be a partition of a set A and let

$$R = \left\{ (a, b) : a, b \in A \text{ and } \exists S \in \mathcal{S} \text{ such that } a, b \in S \right\}.$$

Then R is an equivalence relation on A, and its equivalence classes are exactly the sets of the partition $\mathcal{S}$.

Proof (1/2)

We show that R is an equivalence relation.

- Reflexive: for every $a \in A$ there exists $S \in \mathcal{S}$ with $a \in S$, hence $(a, a) \in R$.
- Symmetric: if $a, b \in S$, the same S contains b and a, hence $(b, a) \in R$.
- Transitive: if $a, b \in S$ and $b, c \in T$ with $S, T \in \mathcal{S}$, then

$$b \in S \cap T \quad \Rightarrow \quad S \cap T \neq \emptyset \quad \Rightarrow \quad S = T,$$

because two distinct sets of the partition are disjoint. Hence $a, c \in S$ and $(a, c) \in R$.

Proof (2/2)

We show that the classes are the sets of the partition. Given $a \in A$, let S_a be the only set of $\mathcal{S}$ containing a.

- If $b \in S_a$, then $(a, b) \in R$, that is $b \in [a]$.
- If $b \in [a]$, there exists $S \in \mathcal{S}$ with $a, b \in S$; since a belongs to only one set of the partition, $S = S_a$ and hence $b \in S_a$.

Therefore $[a] = S_a$: every class is a set of the partition.

Finally, every $S \in \mathcal{S}$ is nonempty: taking $a \in S$, we have $S = S_a = [a]$, hence every set of the partition is a class. □

Example: classes modulo 3 (1/2)

$R_{\equiv_3}$ divides the natural numbers into 3 equivalence classes:

$$[0] = \{0, 3, 6, \dots\}, \qquad [1] = \{1, 4, 7, \dots\}, \qquad [2] = \{2, 5, 8, \dots\}.$$

Each class contains the numbers with the same remainder in the division by 3: if $a = 3\,q_a + r_a$ and $b = 3\,q_b + r_b$, with remainders $r_a, r_b \in \{0, 1, 2\}$,

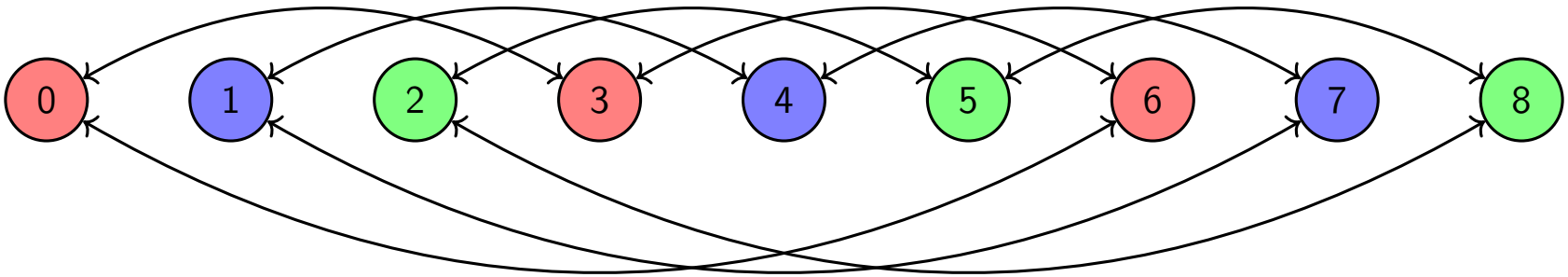
$$b - a = 3\,(q_b - q_a) + (r_b - r_a).$$

- If $r_a = r_b$, then $b - a = 3\,(q_b - q_a)$ and $(a, b) \in R_{\equiv_3}$.
- If $b - a = 3\,q$ with $q \in \mathbb{Z}$, then $r_b - r_a = 3\,(q - q_b + q_a)$ is a multiple of 3 between -2 and 2, hence $r_b - r_a = 0$.

Example: classes modulo 3 (2/2)

The directed graph of the relation on the natural numbers from 0 to 8, with one colour per class.

A double-headed arrow represents the two arcs (a, b) and (b, a); the loops (a, a), present for every a, are not drawn.



Example: even and odd numbers (1/2)

R_{even} divides the natural numbers into 2 equivalence classes:

$$[0] = \{0, 2, 4, 6, \dots\}, \qquad [1] = \{1, 3, 5, 7, \dots\}.$$

If $a = 2\,q_a + r_a$ and $b = 2\,q_b + r_b$, with remainders $r_a, r_b \in \{0, 1\}$, then

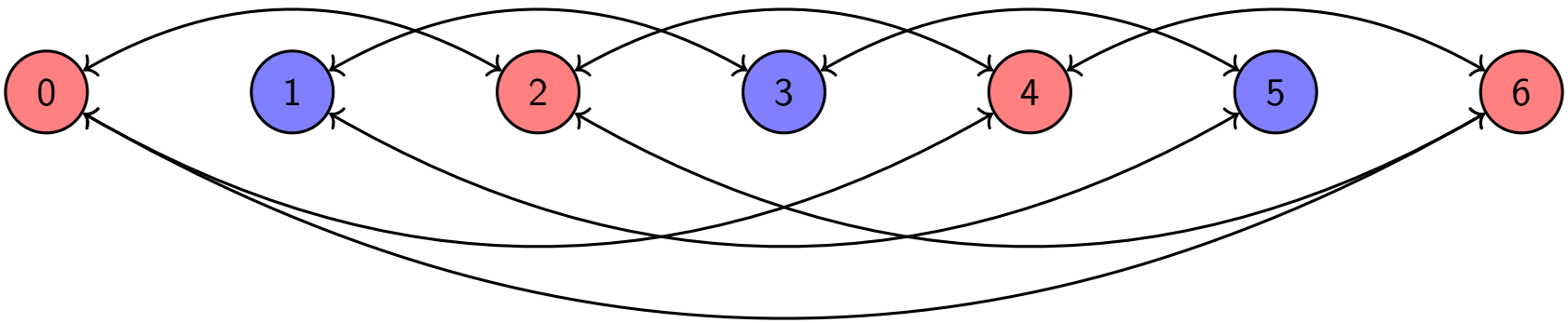
$$a + b = 2\,(q_a + q_b) + (r_a + r_b).$$

- If $r_a = r_b$, then $a + b = 2\,(q_a + q_b + r_a)$ is even.
- If $r_a \neq r_b$, then $a + b = 2\,(q_a + q_b) + 1$ is not even.

Hence $(a, b) \in R_{\text{even}}$ if and only if a and b are both even or both odd.

Example: even and odd numbers (2/2)

The directed graph of the relation on the natural numbers from 0 to 6 (even in red, odd in blue; double-headed arrows and loops omitted as in the previous example):



Equivalence and antisymmetry

Observation 3: equivalence and antisymmetry

If an equivalence relation R on a set A is also antisymmetric, then every equivalence class has a single element: $[a] = \{a\}$ for every $a \in A$.

- In this case $(a, b) \in R$ if and only if $a = b$: the only equivalence relation on A that is also antisymmetric is the relation "equal to".

Proof

Let $a \in A$. By reflexivity $a \in [a]$.

If $b \in [a]$, then

$$\begin{aligned} (a,b) \in R &\Rightarrow (b,a) \in R \quad \text{(symmetry)} \\ &\Rightarrow a = b \quad \text{(antisymmetry).} \end{aligned}$$

Hence $[a] = \{a\}$. □

Summary (1/2)

Equivalence relations:

- Reflexive, symmetric and transitive (e.g. $R_=$, $R_{\equiv_n}$, R_{even}; not $R_<$, $R_\leq$)
- Equivalence class: $[a] = \{b \in A : (a, b) \in R\}$
- $(a, b) \in R \Rightarrow [a] = [b]$

Partitions:

- Family of nonempty, pairwise disjoint sets with union A
- The classes of an equivalence relation form a partition, and every partition is the family of the classes of an equivalence relation

Summary (2/2)

- Classes modulo 3: the numbers with the same remainder in the division by 3
- Classes of R_{even}: the even numbers and the odd numbers
- Equivalence and antisymmetry together: classes with a single element, that is the relation "equal to"

Partial order relations

Introduction: Partial order relations

- In this section we study **reflexive, antisymmetric and transitive** relations: the partial order relations
- We will see that "$\leq$", "$=$", "$\subseteq$", "is a multiple of" and "is a divisor of" are partial order relations
- We will see that the graph of a partial order has no cycles
- We will define **maximum**, **maximal**, **minimum** and **minimal** elements

Partial order relation

Definition 8: of partial order relation

A reflexive, antisymmetric and transitive relation is a ***partial order relation****.*

Example

- The relation $R_{\leq}$ is a partial order relation, but the relation $R_{<}$ is not, since it is not reflexive.
- The relation $R_{=}$ is a partial order relation, because it is reflexive, antisymmetric and transitive.

The relation $R_{\subseteq}$

Observation 4

The relation $R_{\subseteq}$ is a partial order relation

Proof (1/2)

We must prove that the relation is reflexive, antisymmetric and transitive.

- For it to be reflexive we must prove that $(S, S) \in R_{\subseteq}$, which is true since $S \subseteq S$.
- For it to be antisymmetric we must prove that if $S_1 \neq S_2$ then $S_1 \not\subseteq S_2$ or $S_2 \not\subseteq S_1$ or both, the negation of the desired property. Since $S_1 \neq S_2$:

  either there exists some element that is in S_1 but not in S_2, hence $S_1 \not\subseteq S_2$
  or there exists some element that is in S_2 but not in S_1, hence $S_2 \not\subseteq S_1$
  or both of the previous options hold

Proof (2/2)

- For it to be transitive we must prove that $(S_1, S_2) \in R_{\subseteq}$ and $(S_2, S_3) \in R_{\subseteq}$ imply $(S_1, S_3) \in R_{\subseteq}$. Clearly, since $S_1 \subseteq S_2$ and $S_2 \subseteq S_3$, we have $S_1 \subseteq S_3$.

□

Example: "is a multiple of" (1/2)

The relation R_{mul} is a partial order relation.

- Reflexive: $a = 1 \cdot a$, hence $(a, a) \in R_{\mathrm{mul}}$.
- Transitive: if $a = q\,b$ and $b = p\,c$ with $q, p \in \mathbb{N}$, then

$$a = q\,b = q\,(p\,c) = (q\,p)\,c, \qquad \text{with} \quad q\,p \in \mathbb{N},$$

  hence $(a, c) \in R_{\mathrm{mul}}$.
- Antisymmetric: if $a = q\,b$ and $b = p\,a$ with $q, p \in \mathbb{N}$, then

$$a = q\,b = q\,p\,a.$$

Example: "is a multiple of" (2/2)

From $a = q\,p\,a$:

- if $a = 0$, then $b = p \cdot 0 = 0 = a$;
- if $a \neq 0$, dividing by a we obtain $q\,p = 1$.

  Neither factor is 0, hence $q \geq 1$ and $p \geq 1$; if $q \geq 2$ we would have $q\,p \geq 2\,p \geq 2$. Hence $q = 1$, and in the same way $p = 1$.

  Therefore $a = 1 \cdot b = b$.

Example: "is a divisor of"

Also R_{div} is a partial order relation, because $(a, b) \in R_{\mathrm{div}}$ if and only if $(b, a) \in R_{\mathrm{mul}}$:

- reflexive: $(a, a) \in R_{\mathrm{mul}}$, hence $(a, a) \in R_{\mathrm{div}}$;
- transitive: if $(a, b), (b, c) \in R_{\mathrm{div}}$, then $(c, b), (b, a) \in R_{\mathrm{mul}}$, hence $(c, a) \in R_{\mathrm{mul}}$, that is $(a, c) \in R_{\mathrm{div}}$;
- antisymmetric: if $(a, b), (b, a) \in R_{\mathrm{div}}$, then $(b, a), (a, b) \in R_{\mathrm{mul}}$, hence $a = b$.

Partially ordered set

Definition 9: of partially ordered set

*A **partially ordered set** is a pair consisting of a set and a partial order relation defined on it.*

Example

- The sets of natural, rational or real numbers with the relation $R_{\leq}$ are partially ordered sets.
- The relation "is a descendant of" defined on a subset of people is a partial order relation (if we consider individuals as descendants of themselves).
- Consequently, the subset of people considered with the relation "is a descendant of" is a partially ordered set.
- The set of natural numbers with R_{mul}, or with R_{div}, is a partially ordered set.

Graph of a partial order

Observation 5: graph of a partial order

The directed graph of a partial order relation contains no ***cycles****, that is, sequences of arcs*

$$(a_1, a_2),\ (a_2, a_3),\ \ldots,\ (a_{k-1}, a_k),\ (a_k, a_1)$$

with $k \geq 2$ *distinct vertices* $a_1, a_2, \ldots, a_k$*. Once the loops are removed, the graph is* ***acyclic****.*

Proof

Suppose by contradiction that the graph contains a cycle.

- If $k = 2$, the cycle consists of the arcs (a_1, a_2) and (a_2, a_1).
- If $k \geq 3$, by transitivity

$$(a_2, a_3),\ (a_3, a_4) \quad \Rightarrow \quad (a_2, a_4) \in R,$$

$$(a_2, a_4),\ (a_4, a_5) \quad \Rightarrow \quad (a_2, a_5) \in R,$$

and so on up to the arc (a_k, a_1), from which $(a_2, a_1) \in R$.

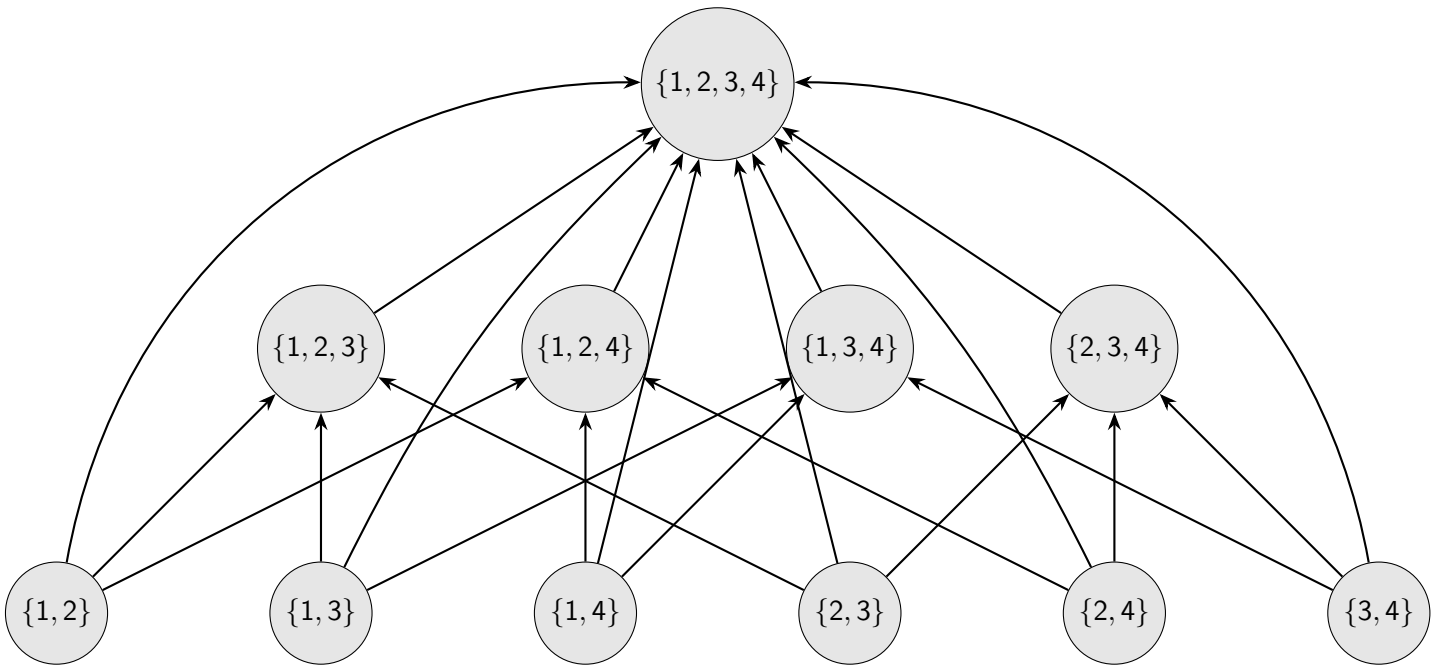
In both cases $(a_1, a_2) \in R$ and $(a_2, a_1) \in R$, and by antisymmetry $a_1 = a_2$, against the assumption that the vertices of the cycle are distinct. □

The converse does not hold

- An acyclic graph does not necessarily represent a partial order relation.
- For example, the relation $\{(1,2),(2,3)\}$ on the set $\{1,2,3\}$ has a graph without cycles, but:
  - it is not reflexive: $(1,1)$ does not belong to the relation;
  - it is not transitive: $(1,3)$ does not belong to the relation.

Example

The graph of $R_{\subseteq}$ on the subsets of $\{1, 2, 3, 4\}$. The empty set, the sets consisting of a single number (and the arcs leaving them) and the loops (S, S) are not shown; all the other arcs are drawn.



In partially ordered sets it makes sense to ask which elements are the "largest" and the "smallest".

Let us now see maximum, maximal, minimum and minimal elements.

Maximum, maximal, minimum and minimal elements

Definition 10: of maximum, maximal, minimum and minimal element

Let (A, R) be a partially ordered set.

- $\tilde{a} \in A$ *is the* ***maximum*** *of A if $(b, \tilde{a}) \in R$ for every $b \in A$.*
- $\bar{a} \in A$ *is a* ***maximal element*** *if there is no $b \in A$ with $(\bar{a}, b) \in R$ and $b \neq \bar{a}$.*
- $\tilde{a} \in A$ *is the* ***minimum*** *of A if $(\tilde{a}, b) \in R$ for every $b \in A$.*
- $\bar{a} \in A$ *is a* ***minimal element*** *if there is no $b \in A$ with $(b, \bar{a}) \in R$ and $b \neq \bar{a}$.*

Maximum and maximal elements on the graph

- The **maximum** is the vertex that receives an arc from every vertex.
- A **maximal element** is a vertex from which no arc leaves, apart from the loop.
- For the **minimum** and the **minimal** elements, incoming and outgoing arcs are swapped.

Uniqueness of the maximum

Proposition 3: uniqueness of the maximum and of the minimum

In a partially ordered set (A, R) the maximum, if it exists, is unique. The same holds for the minimum.

Proof

Let $\tilde{a}_1$ and $\tilde{a}_2$ be two maxima of A:

$$\tilde{a}_2 \text{ maximum} \quad \Rightarrow \quad (\tilde{a}_1, \tilde{a}_2) \in R,$$

$$\tilde{a}_1 \text{ maximum} \quad \Rightarrow \quad (\tilde{a}_2, \tilde{a}_1) \in R.$$

By antisymmetry $\tilde{a}_1 = \tilde{a}_2$.

If $\tilde{a}_1$ and $\tilde{a}_2$ are two minima, then $(\tilde{a}_1, \tilde{a}_2) \in R$ because $\tilde{a}_1$ is a minimum and $(\tilde{a}_2, \tilde{a}_1) \in R$ because $\tilde{a}_2$ is a minimum, and again $\tilde{a}_1 = \tilde{a}_2$. □

The maximum is the only maximal element

Observation 6: the maximum is the only maximal element

If a partially ordered set (A, R) has the maximum $\tilde{a}$, then $\tilde{a}$ is a maximal element, and it is the only maximal element of A.

Proof

- $\tilde{a}$ is maximal: if $(\tilde{a}, b) \in R$, since $\tilde{a}$ is the maximum also $(b, \tilde{a}) \in R$, and by antisymmetry $b = \tilde{a}$.
- $\tilde{a}$ is the only maximal element: let $\bar{a}$ be a maximal element. Since $\tilde{a}$ is the maximum,

$$(\bar{a}, \tilde{a}) \in R;$$

since $\bar{a}$ is maximal, $\tilde{a} \neq \bar{a}$ is impossible. Hence $\bar{a} = \tilde{a}$.

□

The maximum may not exist

- In a partially ordered set the **maximum element** may not exist, that is, there may be no element a such that:

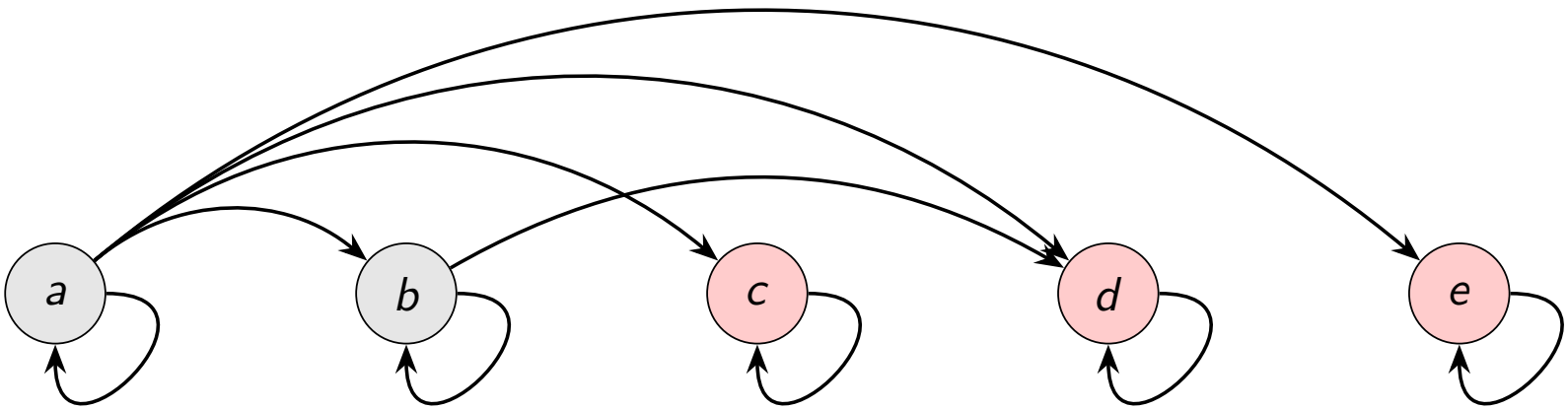
$$\forall b \in A, \qquad (b, a) \in R$$

- A partially ordered set may therefore contain several maximal elements a such that, for no $b \in A$ with $b \neq a$, we have $(a, b) \in R$.

Example: three maximal elements

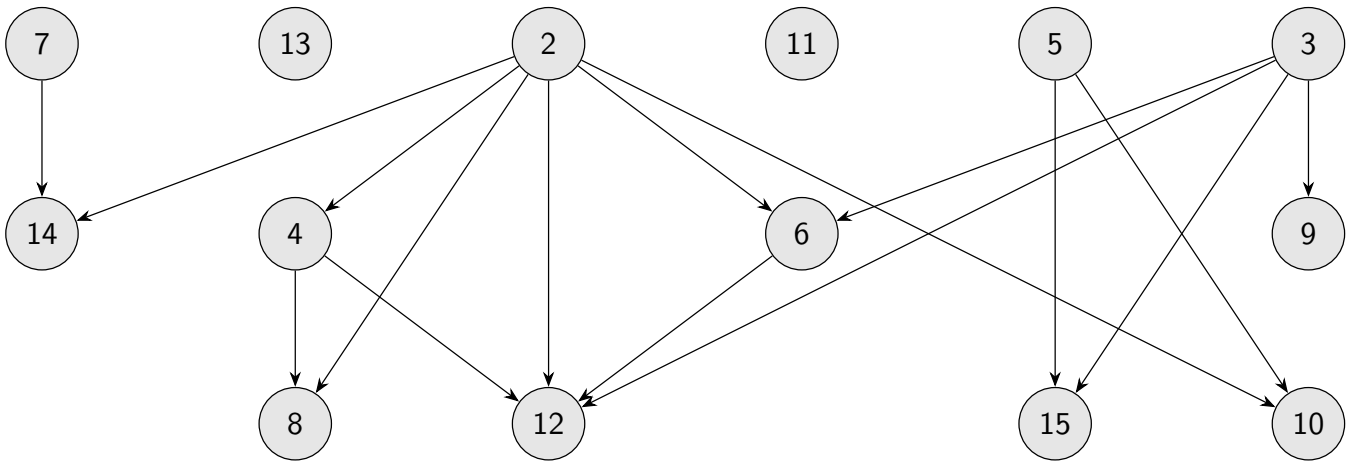
The graph of a partial order relation on $\{a, b, c, d, e\}$ with three maximal elements, c, d and e, in red.

The maximum does not exist: for example (d, c) does not belong to the relation. The element a is the minimum.



Example: divisors in $\{2, 3, \ldots, 15\}$

The graph of "is a divisor of" on $\{2, 3, \ldots, 15\}$ (loops not drawn). Maximal: $8, 9, \ldots, 15$; minimal: $2, 3, 5, 7, 11, 13$; neither the maximum nor the minimum exists.



Example

Given a set of boxes of different sizes, the relation "one box is contained in the other" on the set of boxes may have several maximal boxes, that is, boxes that are not contained in any other box.

Summary (1/2)

Partial order:

- Reflexive, antisymmetric and transitive (e.g. $R_{\leq}$, $R_{=}$, $R_{\subseteq}$, R_{mul}, R_{div}; not $R_{<}$)
- **Partially ordered set**: a set with a partial order relation
- The graph of a partial order, once the loops are removed, is acyclic; the converse does not hold

Summary (2/2)

Maximum and maximal elements:

- Maximum: $(b, \tilde{a}) \in R$ for every b; maximal: no $b \neq \bar{a}$ with $(\bar{a}, b) \in R$
- Minimum and minimal: the same definitions with the pairs swapped
- The maximum, if it exists, is unique and it is the only maximal element
- There may be several **maximal** elements and no maximum

Total order relations

Introduction: Total order relations

- In this section we require that **every pair** of elements be comparable
- We will define total relations, total order relations and totally ordered sets
- The natural, rational and real numbers with "$\leq$" are the fundamental example
- We will see that a totally ordered set has at most one maximal element

Total relation

Definition 11: of total relation

*A relation R on a set A is a **total relation** if:*

$$\forall a, b \in A, \qquad (a, b) \in R \quad \text{or} \quad (b, a) \in R \quad (\text{or both})$$

Observation 7: a total relation is reflexive

Every total relation R on a set A is reflexive.

Proof

Let $a \in A$. Choosing $b = a$ in the definition of total relation we obtain

$$(a, a) \in R \quad \text{or} \quad (a, a) \in R,$$

that is $(a, a) \in R$. □

Example

- The relation $R_{\leq}$ is a total relation.
- The relation $R_{\subseteq}$ is not a total relation: with $S_1 = \{1, 2\}$ and $S_2 = \{2, 3\}$, $(S_1, S_2) \notin R_{\subseteq}$ and $(S_2, S_1) \notin R_{\subseteq}$.
- The relation "is a descendant of" is not a total relation since there exist pairs of individuals (a, b) for which neither a descends from b nor b descends from a.
- The relation $R_{<}$ is not a total relation, because it is not reflexive: for $a = b = 1$ the definition requires $(1, 1) \in R_{<}$, which is false.
- The relation $R_{=}$ is not a total relation: for example $(3, 4) \notin R_{=}$ and $(4, 3) \notin R_{=}$.

Total order relation

Definition 12: of total order relation

A partial order relation that is also a total relation is a ***total order relation****.*

Example

- $R_{\leq}$ is a total order relation.
- $R_{=}$ is a partial order but not a total order, because it is not total.
- $R_{<}$ is not a total order, because it is not a partial order.
- $R_{\subseteq}$ is a partial order but not a total order: $\{1, 2\}$ and $\{2, 3\}$ are not comparable.
- R_{mul} is a partial order but not a total order: $(6, 9) \notin R_{\text{mul}}$, because there is no $k \in \mathbb{N}$ with $6 = k \cdot 9$, and $(9, 6) \notin R_{\text{mul}}$, because there is no $k \in \mathbb{N}$ with $9 = k \cdot 6$.

Example: divisibility among powers of 2

The relation "is a divisor of" on the powers of 2

$$R = \left\{ (a, b) : a, b \in A \ \text{ and } \ \exists k \in \mathbb{N} \ \text{ such that } \ b = k\, a \right\},$$

with $A = \{2^n : n \in \mathbb{N}\}$, is a total order relation.

- It is a partial order: the three properties hold for R_{div} on all the naturals, hence also on the elements of A.
- It is total: let $a = 2^m$ and $b = 2^n$. If $m \leq n$,

$$b = 2^n = 2^{n-m} \cdot 2^m = 2^{n-m}\, a, \qquad \text{with } \ 2^{n-m} \in \mathbb{N},$$

hence $(a, b) \in R$. If $n < m$, in the same way $a = 2^{m-n}\, b$ and $(b, a) \in R$.

Totally ordered set

Definition 13: of totally ordered set

*A **totally ordered set** is a pair consisting of a set and a total order relation defined on it.*

- Example: the sets of natural, rational or real numbers with the relation $R_{\leq}$ are totally ordered sets.

Maximal element and maximum in a total order

Proposition 4: maximal element and maximum in a totally ordered set

In a totally ordered set (A, R) an element is maximal if and only if it is the maximum. Hence a totally ordered set has at most one maximal element.

- "At most" cannot be replaced by "exactly": $(\mathbb{N}, R_{\leq})$ has no maximal elements, because for every $a \in \mathbb{N}$ we have $(a, a+1) \in R_{\leq}$ with $a+1 \neq a$.

Proof

If $\tilde{a}$ is the maximum, then it is maximal (this holds in every partially ordered set).

Conversely, let $\bar{a}$ be maximal and let $b \in A$. Since the relation is total,

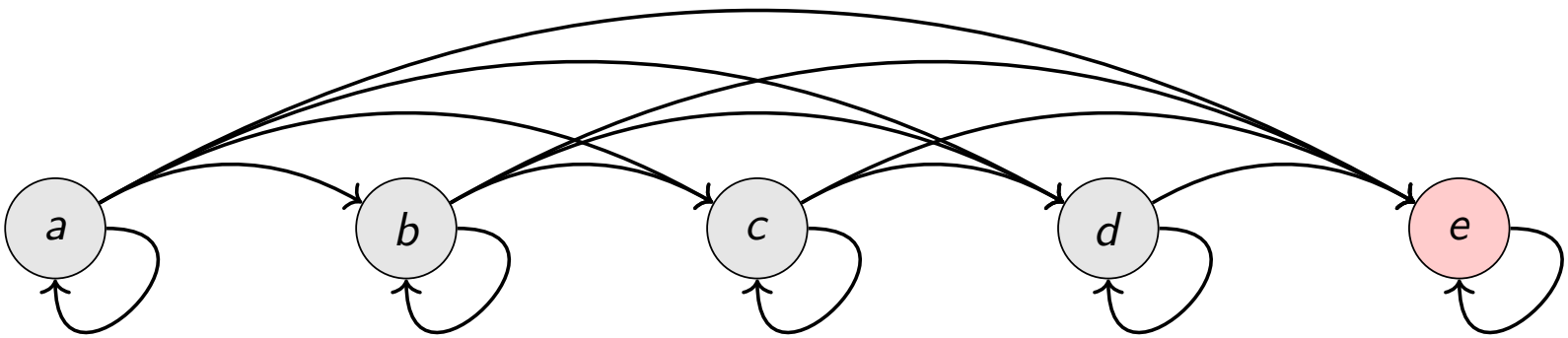
$$(b, \bar{a}) \in R \quad \text{or} \quad (\bar{a}, b) \in R.$$

In the second case, since $\bar{a}$ is maximal, $b = \bar{a}$, and hence $(b, \bar{a}) = (\bar{a}, \bar{a}) \in R$ by reflexivity.

In both cases $(b, \bar{a}) \in R$, hence $\bar{a}$ is the maximum. Since the maximum, if it exists, is unique, there is at most one maximal element. □

Example: maximum

The directed graph of a total order relation on the set $\{a, b, c, d, e\}$, with the maximum e in red:



Summary

- **Total relation**: for all a, b we have $(a, b) \in R$ or $(b, a) \in R$; every total relation is reflexive (hence $R_<$ is not total)
- **Total order**: partial order that is also total (e.g. $R_\leq$; "is a divisor of" on the powers of 2)
- $R_=$, $R_\subseteq$, R_{mul} and "is a descendant of" are not total
- $\mathbb{N}$, $\mathbb{Q}$, $\mathbb{R}$ with $\leq$ are **totally ordered sets**
- In a totally ordered set maximal element and maximum coincide: there is **at most** one maximal element

Functions

Introduction: Functions

- In this section we see **functions** as particular binary relations
- We will define domain, codomain, argument and value
- Functions are the central object of the whole Mathematical Analysis course

Function

Definition 14: of function

Given two sets A and B, a binary relation f on A and B is a ***function*** *if, for each $a \in A$, there exists one and only one $b \in B$ such that $(a, b) \in f$.*

Definition 15: of domain and codomain

The set A is called the ***domain*** *of f, and the set B is called the* ***codomain*** *of f.*

Notation

- We write:

$$f : A \to B$$

and if $(a, b) \in f$, we write:

$$b = f(a)$$

since b is uniquely determined by the choice of a.

- Intuitively, the function f assigns an element of B to each element of A. No element of A is associated with two different elements of B. The same element of B can however be assigned to different elements of A.

Example

- The binary relation on $\mathbb{N}$ and $\{0,1\}$:

$$f = \Big\{(a,b) : a \in \mathbb{N},\ b \in \{0,1\} \ \text{ and } \ b = a \mod 2\Big\},$$

where $a \mod 2$ is the remainder of the division of a by 2, *is a function* $f : \mathbb{N} \to \{0,1\}$ since for all natural numbers a, there is exactly one value $b \in \{0,1\}$ such that $b = a \mod 2$. For example,

$$0 = f(0), \quad 1 = f(1), \quad 0 = f(2), \ldots$$

Example

- The binary relation

$$g = \left\{ (a, b) : a, b \in \mathbb{N} \quad \text{and} \quad a + b \text{ is even} \right\}$$

is not a function, since for example (1, 3) and (1, 5) are both in g. In other words, for $a = 1$, we do not have one and only one b such that $(a, b) \in g$.

Argument and value

Definition 16: of argument and value

Given a function $f : A \to B$, if $b = f(a)$, we say that $a \in A$ is the argument of f and that $b \in B$ is the value of f associated with a.

- We can **define a function** by *directly defining its value* for all the *elements* of its *domain*.

Example

For example, we can define $f(n) = 2\,n$ for $n \in \mathbb{N}$, which means:

$$f = \{(n, 2\,n) : n \in \mathbb{N}\} \quad \textit{and} \quad f : \mathbb{N} \to \mathbb{N}$$

Summary

- A **function** $f : A \to B$ is a binary relation in which each $a \in A$ is related to one and only one $b \in B$
- A is the **domain**, B the **codomain**; we write $b = f(a)$
- a is the **argument**, b the **value** of f at a
- $b = a \bmod 2$ is a function; "$a + b$ even" is not
- A function can be defined by assigning the value to each element of the domain

Summary (1/2)

Binary relations:

- Subsets of a Cartesian product $A \times B$, representable with tables and directed graphs
- Properties: reflexive, symmetric, transitive, antisymmetric

Equivalence relations:

- Reflexive, symmetric and transitive (e.g. $=$, congruence modulo n)
- The equivalence classes form a partition, and conversely

Summary (2/2)

Order relations:

- Partial order: reflexive, antisymmetric, transitive (e.g. $\leq$, $\subseteq$, "is a multiple of")
- Total order: partial order in which every pair is comparable (e.g. $\leq$)
- Maximum (unique, if it exists), maximal, minimum and minimal elements

Functions:

- Relations that associate with each element of the domain one and only one element of the codomain
- Notation $f : A \to B$, $b = f(a)$